

Arpan Pathak

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📄 SUMMARY

Senior Software Engineer, **8 years**: developer tools, observability platforms, performance-critical distributed systems at **Amazon, Microsoft, Oracle**. Full-stack **Rust, C++, Kotlin, Go, Python, JavaScript, Kubernetes, eBPF/Cilium**, GPU/**CUDA**, cloud **performance optimization**.

🏢 EXPERIENCE

Software Developer II, Microsoft, Redmond, WA

Jun 2025 – Aug 2026

- **Zero-Trust Sovereign Cloud Networking**: eBPF + Cilium policies enforcing **data residency** & **SecNumCloud** compliance; Kubebuilder operators reconciling policy state across **AKS** clusters (France, Germany).
- **C++/C# → Rust Migration (Microsoft Security)**: migrated **Microsoft Information Protection** from unsafe C++/C# to Rust; killed use-after-free/buffer-overflow/null bugs, ~ **40% lower P99**, **2x throughput**, **3x lower memory**, 6-figure savings.
- **AI Policy Engine**: auto-generates Cilium policies, scans misconfigurations; review **4 hours** → **near-real-time**, zero drift.
↳ TechStack: **Rust, C++, eBPF, Cilium, Kubernetes (AKS), Kubebuilder, Prometheus, Linux**

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 – Mar 2025

- **High-Performance Database Engine**: lock-free data structures cutting contention in high-concurrency read-write paths; **45,000 ops/s** throughput gain.
- **Infrastructure-as-Code SDK**: Terraform provider (developer SDK) for Oracle Autonomous Database; **\$1.2M/month** projected savings.
- **Security & Key Management**: OCI Security Vault, automated key rotation across control plane services.
↳ TechStack: **Java, Go, Terraform, OCI, Autonomous Database**

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 – Oct 2024

- **Real-Time ML Ranking Engine**: XGBoost + Deep Learning ensemble across **FreeVee, Twitch, MiniTV, Prime Video, Amazon Retail**; **5% CTR lift**, **\$5.4M/month** revenue.
- **BERT NLP Service**: **5,000 QPS** at peak; triage SLA **7 days** → **2 hours**.
- **Ads Publisher Platform (Event-Driven)**: event-driven systems delivering **programmatic guaranteed ad deals** via real-time **CDC** at **< 50ms** latency (Kotlin, CSP).
- **NoSQL SDK**: two-phase commit, **serializable isolation** at **3K writes/s**.
- **Amazon WorkEvents Data Platform**: **1-petabyte** warehouse/lake with **AWS Glue (Spark)** for financial reporting & pay computation.
↳ TechStack: **Kotlin, Java, Python, PyTorch, XGBoost, Spark, Kafka, DynamoDB, Redshift, AWS**

SDE, Razorpay, Bengaluru, India

Aug 2020 – Feb 2021

- Go UPI gateway: **1M+ txns/hr**, **99.99% uptime**, **150ms P99** at **50K TPM**.

Software Engineer, Mindfire, Bhubaneswar, India

Aug 2018 – Feb 2020

- Java/Spring full-stack + AR search (**10K+ daily queries**); latency **5 min** → **1 min**.

📄 PERSONAL PROJECTS

🚗 Driving-CivicSense: Edge AI Vision System (Rust)

- On-device perception in **Rust** on the **NVIDIA Jetson Orin Nano Super** (67 INT8 TOPS): **YOLOv8/v11 INT8 ONNX** (~ **12 ms**) + Deep SORT/Kalman + proven kinematic engine; **100%** on-device.
- Research paper: **Deterministic Intersection Blockage Prediction** | arpanpathak.github.io/driving-civicsense-vision-model/ | Code: github.com/arpanpathak/driving-civicsense-vision-model
- Books: **Seeing Machines** | arpanpathak.github.io/seeing-machines-book · **CUDA Kernels** (CUDA-Oxide / Rust-to-CUDA) | arpanpathak.github.io/gpu-parallel-book

🔧 SKILLS

↳ **Languages**: Rust, C++, Kotlin, Go, C, Java, Python, TypeScript, JavaScript, Shell Script

🔧 **Systems Programming & Performance**: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

🌐 **Networking & Cloud Security**: TCP/IP, UDP, TLS, HTTP/2/3, QUIC, RDMA, InfiniBand, DPDK, eBPF, XDP, Cilium, VPC, NAT, VPN, Load Balancing

📡 **Distributed Systems & Streaming**: Consensus, Replication, Sharding, Kafka, Flink, Spark, Spark Streaming, Kinesis, CDC, Microservices; gRPC, REST, Protobuf

☁ **Cloud & Infra**: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio

🔗 **ML Inference & AI**: Deep Learning, Computer Vision, PyTorch, TensorFlow, XGBoost, BERT, LLMs, ONNX Runtime, YOLOv8/v11, CNN, NLP, Edge Inference, MLOps

🔧 **GPU & Parallel Computing**: CUDA, CUDA-Oxide (Rust-to-CUDA), SIMT, GPU Kernel Programming

📊 **Observability**: Prometheus, Grafana, Azure Monitor, AWS CloudWatch, Trace & Sampling Data

🗄 **Data & Storage**: DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

🔧 **Core CS**: Data Structures & Algorithms, System Design, Operating Systems, Linux

🎓 EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018